

Transforming Quadratic Functions to Vertex Form

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The **vertex form** of a quadratic function is

$$y = a(x - h)^2 + k.$$

Vertex

The **vertex** is the point (h, k) .
For

$$y = 2(x - 3)^2 + 4$$

the vertex is

$$(3, 4).$$

Value of a

The value of a determines the **opening direction** and the **width** of the parabola.

If $a > 0$, it opens upward.

If $a < 0$, it opens downward.

Already in Vertex Form

If the equation is already in the form

$$y = a(x - b)^2$$

then simply write it as

$$y = a(x - b)^2 + 0.$$

Basic Quadratic Forms

For equations of the form $y = ax^2$ or $y = ax^2 + b$, write them as

$$y = a(x - 0)^2 + k.$$

For example,

$$y = 3x^2 - 4$$

becomes

$$y = 3(x - 0)^2 - 4.$$

Completing the Square

For equations of the form

$$y = ax^2 + bx + c$$

use the method of **completing the square**.

Steps in Completing the Square

To complete the square:

Divide the coefficient of x by 2.

Square the result.

Add and subtract this value.

Rewrite the perfect square trinomial as $(x - h)^2$.

Examples

Example 1

Transform

$$y = 2x^2$$

into vertex form.

$$y = 2(x - 0)^2 + 0$$

Example 2

Transform

$$y = -5x^2 + 4$$

into vertex form.

$$y = -5(x - 0)^2 + 4$$

Example 3

Transform

$$y = 3(x - 2)^2$$

into vertex form.

$$y = 3(x - 2)^2 + 0$$

Example 4

Transform

$$y = -4(x + 1)^2$$

into vertex form.

$$y = -4(x + 1)^2 + 0$$

Example 5

Transform

$$y = x^2 + 8x + 10$$

into vertex form.

$$y = (x + 4)^2 - 6$$

Example 6

Transform

$$y = x^2 - 12x + 20$$

into vertex form.

$$y = (x - 6)^2 - 16$$

Example 7

Transform

$$y = 2x^2 + 12x + 5$$

into vertex form.

$$y = 2(x + 3)^2 - 13$$

Example 8

Transform

$$y = 3x^2 - 18x + 6$$

into vertex form.

$$y = 3(x - 3)^2 - 21$$

Example 9

Transform

$$y = -x^2 + 4x - 7$$

into vertex form.

$$y = -(x - 2)^2 - 3$$

Example 10

Transform

$$y = -2x^2 - 8x + 3$$

into vertex form.

$$y = -2(x + 2)^2 + 11$$

Practice Exercises

Practice 1

Transform

$$y = 2x^2$$

into the form $y = a(x - h)^2 + k$.

Practice 2

Transform

$$y = -5x^2 + 4$$

into the form $y = a(x - h)^2 + k$.

Practice 3

Transform

$$y = 3(x - 2)^2$$

into the form $y = a(x - h)^2 + k$.

Practice 4

Transform

$$y = -4(x + 1)^2$$

into the form $y = a(x - h)^2 + k$.

Practice 5

Transform

$$y = x^2 + 8x + 10$$

into the form $y = a(x - h)^2 + k$.

Practice 6

Transform

$$y = x^2 - 12x + 20$$

into the form $y = a(x - h)^2 + k$.

Practice 7

Transform

$$y = 2x^2 + 12x + 5$$

into the form $y = a(x - h)^2 + k$.

Practice 8

Transform

$$y = 3x^2 - 18x + 6$$

into the form $y = a(x - h)^2 + k$.

Practice 9

Transform

$$y = -x^2 + 4x - 7$$

into the form $y = a(x - h)^2 + k$.

Practice 10

Transform

$$y = -2x^2 - 8x + 3$$

into the form $y = a(x - h)^2 + k$.

Activities

Activity 1

Transform

$$y = 7x^2$$

into the form $y = a(x - h)^2 + k$.

Activity 2

Transform

$$y = -3x^2 - 6$$

into the form $y = a(x - h)^2 + k$.

Activity 3

Transform

$$y = 5(x + 4)^2$$

into the form $y = a(x - h)^2 + k$.

Activity 4

Transform

$$y = -2(x - 6)^2$$

into the form $y = a(x - h)^2 + k$.

Activity 5

Transform

$$y = x^2 + 10x + 1$$

into the form $y = a(x - h)^2 + k$.

Activity 6

Transform

$$y = x^2 - 4x + 9$$

into the form $y = a(x - h)^2 + k$.

Activity 7

Transform

$$y = 2x^2 - 16x + 7$$

into the form $y = a(x - h)^2 + k$.

Activity 8

Transform

$$y = 4x^2 + 24x - 8$$

into the form $y = a(x - h)^2 + k$.

Activity 9

Transform

$$y = -x^2 - 10x + 6$$

into the form $y = a(x - h)^2 + k$.

Activity 10

Transform

$$y = -3x^2 + 18x - 5$$

into the form $y = a(x - h)^2 + k$.

Thank you!